Johnathan W. Carroll

Infrastructure, Security & AI Platform Engineering Leader

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Online resume / Cover letter / Research

Hands-on technology leader with 25+ years across government, enterprise IT, ISP operations, entrepreneurship, and applied AI. Combines systems architecture and operational leadership with infrastructure modernization, cybersecurity, disaster recovery, AI infrastructure, and LLMOps.

Current Experience

Pascua Yaqui Tribe Government

Tucson, Arizona

IT Infrastructure Manager — Aug 2023 - Present
Senior System Administrator — Jan 2021 - Aug 2023
Security Analyst — Jan 2019 - Jan 2021

- Lead a nine-person Infrastructure and SecOps organization supporting approximately 1,500 employees across seven sites and critical government services.
- Serve as the sole systems infrastructure engineer and systems/security architect, reporting to the IT Director; own compute, virtualization, storage, capacity planning, technical standards, and lifecycle strategy.
- Modernized and consolidated legacy platforms around Nutanix AHV, Cisco UCS, and Pure Storage, producing approximately $250,000 in annual operating savings and a 50% reduction in physical infrastructure footprint.
- Lead infrastructure vendor evaluation, technical requirements, RFPs, procurement recommendations, licensing, and renewals; develop and manage infrastructure portions of the annual CapEx and OpEx budget.
- Led Darktrace adoption and expanded layered security through Microsoft Defender, Sentinel, Rapid7 MDR, vulnerability management, managed patching, and employee awareness. Helped establish dedicated Cybersecurity and Physical Security functions.
- Designed and maintain backup and disaster-recovery architecture using Veeam, immutable backups, storage replication, and geographically separated recovery infrastructure.
- Originated and serve as primary architect for an estimated $6 million centralized data-center and combined SOC/NOC program, including APC POD, power, cooling, civil infrastructure, and migration planning. The partially funded program remains in design and procurement, not commissioned.
- Translate independent AI research into the emerging Tribal AI architecture, governance, policy, and security program, including collaboration with the Tribal Attorney General. The program remains in development and scoped pilot testing.

TechSoft Systems

Founder, Owner & Operator 2012 - Present

- Founded and operate a technology consultancy, now focused primarily on independent applied AI R&D alongside the full-time Tribal role; specialize in small local AI labs and custom GPU/server systems.
- Engineered a Threadripper PRO lab around dual RTX PRO 6000 Blackwell GPUs with 192 GB aggregate GPU memory, 512 GB ECC DDR5 memory, and PCIe 5.0 NVMe storage. Dedicated AI research began in January 2025.
- Convert base models to NVFP4 with NVIDIA Model Optimizer, then test, tune, and deploy with vLLM. LLMOps work spans CUDA, FP8 KV caching, speculative decoding, shared-base LoRA serving and adapter lifecycle validation, embeddings/reranking, multimodal services, observability, and reproducible container operations. Use Ollama for basic testing and selected project embedding workloads.
- Engineered vLLM/NVFP4 residency workflows, recording 0.96-0.97-second source-sleep-to-first-visible-token latency in selected lab tests. A separate historical Lightning runtime candidate passed 61 sleep/wake cycles and 122/122 post-wake arithmetic/tool checks, with approximately 473 ms mean wake time. Results qualify residency components, not the complete agent architecture.
- Diagnosed and corrected an inference proxy connection mismatch; a subsequent approximately 30-minute direct-proxy soak completed 2,648/2,648 requests. Applied controlled comparisons, output validation, and rollback procedures to vLLM deployment changes.
- Evaluated four Qwen/Nemotron NVFP4 embedding checkpoints on a common vLLM runtime, comparing serving performance and retrieval quality across three public datasets; validated 20,480 requests across four 60-second, eight-client tests with zero reported errors.

Selected AI Research

Dynamic GPU Residency & LLMOps

Historical lab tests; residency components validated

Designed a multi-model, multi-agent inference architecture; validated residency components under embedding and host-memory bandwidth load with reproducible timing, correctness checks, fault telemetry, and rollback records.

Historical component validation, not whole-architecture production qualification. Nano numerical failures and long-duration near-saturation testing remained unresolved in the record.

NVIDIA AI-Q Research Assistant

Implemented private research platform; evaluation ongoing

Built a self-hosted AI-Q/NeMo platform and reproducible evidence evaluation; repaired a specific citation-accounting defect, validated with 196 focused passing tests.

PIPER Intent Gateway

Implemented advisory lab integration; historical regression coverage

Architected a Hermes advisory intent/memory plugin with deterministic contracts, monitoring, human review, historical regression coverage, and a selected 106-decision audit.

Previous Experience

Tucson Electric Power

Security Analyst - Contract 2017 - 2019

- Led design, modernization, and integration of CCTV, access control, and physical-security technology into IT across headquarters, power generation, and Arizona substations.
- Contributed security engineering within a $40M+ modernization program, including a $30M data-center conversion contained within that total; contracted through PrevenTronics and later Tech Systems.

TBM Equities LLC / Sahara Apartments

IT Director / Senior System Administrator Jul 2013 - Jul 2018

- Initially engaged through TechSoft following ransomware; restored affected services and business operations within 24 hours, establishing a multi-year IT leadership relationship.
- Reduced hardware footprint 60% with Hyper-V and annual energy costs approximately $8,000; implemented Veeam/off-site replication with a 15-minute RPO for critical systems and networks serving 900+ residents.

BestComm Networks

Technical Engineer / Supervisor 2006 - 2012

- Provided advanced Tier 3 engineering and 24/7 NOC support within an international Best Western provider environment of approximately 3,400 properties; supervised staff and supported network modernization, monitoring, and complex escalations.

TheRiver ISP

Technical Manager 2004 - 2006

- Managed a 52-person technical operations organization serving approximately 26,000 subscribers; led staffing, escalations, carrier relationships, capacity planning, modernization, and the technical/workforce transition following acquisition.

Microsoft Support Contracts

Advanced Technical Support & Escalation Approximately 2000 - 2004

- Delivered Microsoft-authorized third-party support through Keen and later Convergys, with Microsoft-provided training: final-tier server case ownership in the earlier assignment and later Tier 2/3 Windows escalation and mentoring.

OneSmack.com LLC

Founder, Owner & Operator Mid-1990s - 2000 | Norfolk, Virginia

- Founded a web-development and technology business serving approximately 26 clients; delivered database-backed applications, onboarding/quoting workflows, hosting administration, custom systems, and office IT integration.

Technical Domains

Infrastructure	Nutanix AHV, VMware, Hyper-V, Cisco UCS, Pure Storage, NetApp, Veeam, Linux, GPU/server engineering
Networking	Cisco Nexus/Catalyst, Palo Alto, VLAN/VXLAN, VPN, multi-site infrastructure
Security	Darktrace, Defender, Sentinel, Rapid7, Syxsense, identity controls, incident response, vulnerability management
AI Infrastructure & LLMOps	CUDA, NVIDIA Model Optimizer, NVFP4, vLLM, GPU residency, capacity planning, inference evaluation, embeddings/reranking, AI-Q, NeMo Agent Toolkit, Hermes, Honcho, MCP
Engineering	Docker, Python, Bash, PowerShell, SQL, TypeScript/JavaScript, FastAPI, PostgreSQL, Redis, GitHub Actions
Observability	Prometheus, Grafana, Loki, NVIDIA DCGM Exporter, infrastructure and GPU telemetry

Education & Affiliations

- Computer Programming & Digital Repair Certificates - Norfolk Technical Vocational Center
- Coursework - Tidewater Community College
- InfraGard Arizona Member; open-source contributor and advocate